Arista Access Node WFD-ACC-14 Standard Operating Procedure for BGP Route Churn, OSPF Adjacency Loss, and ISIS LSP Flooding Alarm ALARM_1244 at Whitefield POP

Node: WFD-ACC-14 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_1244

Description

This Standard Operating Procedure (SOP) details the troubleshooting and resolution steps for a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) failure domain on the Arista Access Node WFD-ACC-14 at the Whitefield POP. The failure domain is characterized by a high frequency of BGP route advertisements, OSPF adjacency flaps, and ISIS LSP flooding, leading to a warning severity alarm (ALARM_1244). This phenomenon is primarily driven by the interaction between the BGP, OSPF, and ISIS routing protocols, which can cause instability and oscillations in the routing table. The underlying hardware mechanics, such as the Arista 7280SR switch's advanced forwarding capabilities and the use of 100G Ethernet links, can also contribute to the complexity of the failure domain. Furthermore, the configuration of the access node, including the routing protocol settings and the use of features like BGP route reflection and OSPF area hierarchy, can also impact the behavior of the routing protocols and contribute to the failure domain. To resolve this issue, it is essential to perform a thorough analysis of the routing protocols and the network topology, identifying the root causes of the instability and implementing targeted corrective actions to mitigate the effects of the failure domain.

Pre-requisite Checks

- check: Verify the redundant link status between the access node WFD-ACC-14 and the core node at Whitefield POP, result: The redundant link must be operational and passing traffic

- check: Confirm the dispatch of remote hands to the Whitefield POP, result: Remote hands must be available to perform physical actions, if required

- check: Review the network topology and routing protocol configuration, result: A clear understanding of the network topology and routing protocol configuration is essential for troubleshooting

Potential Root Causes

Cause	Probability
BGP route churn caused by an unstable BGP neighbor	High
OSPF adjacency loss due to a misconfigured OSPF area hierarchy	Medium
ISIS LSP flooding caused by a faulty link or a misconfigured ISIS authentication	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP table version is 1, main routing table version is 1 BGP router identifier 10.10.10.1, local AS number 64512 Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State/PfxRcd 192.168.1.1 4 64512 1024 1024 1 0 0 00:00:01 1/1 192.168.2.2 4 64512 1024 1024 1 0 0 00:00:01 1/1, Description: This command displays a summary of the BGP neighbor relationships, including the number of messages received and sent, and the number of prefixes received from each neighbor.

Command: show ospf neighbors, Expected Output: OSPF neighbor status Neighbor ID Pri State Dead Time Address Interface 192.168.1.1 1 FULL/DR 00:00:37 10.10.10.2 Eth1 192.168.2.2 1 FULL/DR 00:00:37 10.10.10.3 Eth2, Description: This command displays a summary of the OSPF neighbor relationships, including the neighbor ID, priority, state, and dead time.

Command: show isis adjacency, Expected Output: ISIS adjacency status Neighbor ID Pri State Dead Time Address Interface 192.168.1.1 1 FULL/DR 00:00:37 10.10.10.2 Eth1 192.168.2.2 1 FULL/DR 00:00:37 10.10.10.3 Eth2, Description: This command displays a summary of the ISIS adjacency relationships, including the neighbor ID, priority, state, and dead time.
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Resolution Steps

Step 1: Step: Stabilize the BGP neighbor, Command: clear bgp neighbor 192.168.1.1, Description: This command clears the BGP neighbor relationship with the unstable neighbor 192.168.1.1, which may help to stabilize the routing table.

Step 2: Step: Verify the OSPF area hierarchy configuration, Command: show ospf database database, Description: This command displays the OSPF database, which can be used to verify the OSPF area hierarchy configuration and identify any potential issues.

Step 3: Step: Verify the ISIS authentication configuration, Command: show isis authentication, Description: This command displays the ISIS authentication configuration, which can be used to verify that the authentication is correctly configured and not causing any issues.

Post-Check Verification

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Command: show bgp summary, Expected Output: BGP table version is 1, main routing table version is 1 BGP router identifier 10.10.10.1, local AS number 64512 Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State/PfxRcd 192.168.1.1 4 64512 1024 1024 1 0 0 00:00:01 1/1 192.168.2.2 4 64512 1024 1024 1 0 0 00:00:01 1/1, Description: This command displays a summary of the BGP neighbor relationships, which can be used to verify that the BGP neighbor relationship with the unstable neighbor 192.168.1.1 has been stabilized.

Command: show ospf neighbors, Expected Output: OSPF neighbor status Neighbor ID Pri State Dead Time Address Interface 192.168.1.1 1 FULL/DR 00:00:37 10.10.10.2 Eth1 192.168.2.2 1 FULL/DR 00:00:37 10.10.10.3 Eth2, Description: This command displays a summary of the OSPF neighbor relationships, which can be used to verify that the OSPF adjacency with the unstable neighbor 192.168.1.1 has been established.

Command: show isis adjacency, Expected Output: ISIS adjacency status Neighbor ID Pri State Dead Time Address Interface 192.168.1.1 1 FULL/DR 00:00:37 10.10.10.2 Eth1 192.168.2.2 1 FULL/DR 00:00:37 10.10.10.3 Eth2, Description: This command displays a summary of the ISIS adjacency relationships, which can be used to verify that the ISIS adjacency with the unstable neighbor 192.168.1.1 has been established.
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Roll-back Procedures

- Step: Restore the original BGP neighbor relationship, Command: show bgp neighbor 192.168.1.1, Description: This command displays the original BGP neighbor relationship with the unstable neighbor 192.168.1.1, which can be used to restore the original configuration.

- Step: Restore the original OSPF area hierarchy configuration, Command: show ospf database database, Description: This command displays the original OSPF database, which can be used to restore the original OSPF area hierarchy configuration.

- Step: Restore the original ISIS authentication configuration, Command: show isis authentication, Description: This command displays the original ISIS authentication configuration, which can be used to restore the original configuration.